

Cubaka Bisimwa Gérard

Goma, D.R,Congo | gerardcubakabisimwa@gmail.com | +243 991 102 448 |

[LinkedIn](#) | [GitHub](#)

Profile

Exploration Geologist specialized in geospatial analysis, remote sensing, and data-driven mineral targeting. Experienced in integrating geological field data, GIS, and satellite imagery to identify mineralization zones, with a focus on copper and gold systems in the DRC.

Hands-on experience from mining environments including Twangiza Mining (gold) and Tenke Fungurume Mining (copper), combined with advanced GeoAI project development (GeoCongo AI) for mineral prospectivity mapping.

Strong ability to bridge field geology and modern data-driven exploration approaches.

Skill Set

- **Geoscience & Exploration:** Mineral exploration (Copper, Gold systems), Structural geology & mineralization controls, Geological mapping & field data collection, Ore deposit models.
 - **Remote Sensing & GIS:** QGIS, ArcGIS, Satellite magery analysis (Sentinel, Landsat), Spatial data integration & interpretation.
 - **GeoAI & Data Science:** Geospatial data analysis, Machine learning for mineral targeting, Geological dataset integration, GeoAI workflows (Prithvi, SAM, LandCover).
 - **Field & Mining Operations:** Sampling & logging, Exploration workflows, Working in remote mining environments.
 - **Collaboration & Communication:** Project management, government & NGO partnerships, multilingual communication (English, French, Swahili).
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Professional Experience

GeoAI Developer & Exploration Geologist – GeoCongo AI | Goma, DRC

July 2025 - *Present*

Developed a geoscience-focused AI plateform of RAG chatbot, semantic search with a Vector Supabase database of geological PDFs, and a geospatial dashboard for remote sensing detections.

- Designed and developed **GeoAI workflows for mineral prospectivity mapping**, integrating satellite imagery, geological maps, and geospatial datasets
- Applied **remote sensing techniques** (Sentinel, Landsat) to detect **alteration zones**, structural **lineaments**, and surface indicators of **mineralization**.
- Integrated multi-source datasets (**geology, geophysics, remote sensing**) to improve targeting accuracy in mineral exploration
- Built and implemented **machine learning models** on Google Vertex AI plateforme to support land cover classification and **geological feature extraction**
- Contributed to the formulation of **data-driven** exploration hypotheses and targeting strategies, aligned with modern mineral exploration practices
- Developed automated pipelines for **satellite data processing and geospatial analysis**, improving efficiency and scalability

Exploration Geologist Intern – Tenke Fungurume Mining (Copper) | Lualaba, DRC

2023 - 2024

- Participated in copper exploration programs within the Central African Copperbelt, contributing to field-based geological investigations
- Conducted detailed **geological mapping and lithological logging**, identifying key rock units and alteration patterns associated with copper mineralization

- Assisted in **core logging and sampling activities**, ensuring accurate recording of geological, structural, and mineralization data
- Supported the interpretation of **stratigraphy and structural controls** influencing ore distribution
- Collaborated with senior geologists to integrate field observations into exploration models and target generation
- Contributed to daily field reports and maintained high-quality geological data for further analysis

Geologist & Research Consultant – iTSCi & Division Provinciale des Mines, South Kivu | DRC 2019 – 2022

- Conducted field-based geological investigations in mining regions, contributing to mineral resource understanding and traceability initiatives.
- Collected and analyzed geological data related to mineral occurrences, supporting exploration and regulatory activities.
- Contributed to the development of **technical reports and geological assessments** used by stakeholders in the mining sector
- Supported implementation of **mineral traceability systems**, improving transparency and compliance in supply chains
- Collaborated with government agencies, field teams, and mining stakeholders in complex operational environments
- Applied geospatial analysis techniques to support geological interpretation and decision-making

Geology Intern – Twangiza Mining (Gold) Sud-Kivu | Bukavu, DRC 2018 – 2020

- Supported gold exploration and production-related geological activities in an active mining environment
- Conducted **field observations, mapping, and sampling** to identify mineralized zones and alteration signatures
- Assisted in **orebody interpretation** through analysis of lithology and structural features
- Participated in **sample collection and preparation workflows**, ensuring adherence to quality control procedures
- Contributed to geological data recording and reporting, supporting ongoing exploration decision-making
- Worked closely with field teams under challenging conditions, demonstrating adaptability and strong field discipline

Education

- **Bachelor's degree in Geology (Exploration & Mining Geology)** – Université Officielle de Bukavu, 2019
- **Career-related training of Computing & Python and JavaScript Programming** – Kadea Academy (Kinshasa) & Simplon (Paris)
- **Certificates** – Esri Data Science | AI & Machine Learning | Automation (n8n, Maker)

Languages

- **French:** Native
- **English:** Professional proficiency



- **Swahili:** Native
- **Lingala:** Fluent



Web site and Social Media

- **Portfolio:** <https://ger-cub.github.io/portfoliogerard/>
- **GitHub:** <https://github.com/Ger-Cub/>
- **Linkedin:** <https://www.linkedin.com/in/g%C3%A9rard-cubaka-38a27619b/>

Key Project

- **GeoCongo AI:** <https://www.geocongoai.com>
- **GeoCongo AI GeoSpatial API:** <https://geocongo-solafune-greenfield-api.geocongoai.com>